

Midterm 1

Applied Algebra

April 15, 2025

Name: _____

ID #: _____

Instructions:

- No notes or books are allowed. Calculators are permitted.
- You must show your reasoning, not just the final answer. Answers without justification will receive only partial credit.
- If a problem asks you to solve an equation, you must find all solutions. If no solutions exist, you must explain why.
- Please cross out any work that is not part of your final solution (e.g., preliminary computations you didn't use).

Each problem is worth **25 points**.

	1	2	3	4	Total
Grade					

Exercise 1. (True/False) Justify your answers with a few sentences.

1. For any positive integers a , b , and c , if $a \mid c$ and $b \mid c$, then $ab \mid c$.
2. For any prime number $p \geq 3$, the equation $x^2 = [1]_p$ has exactly two solutions in $\mathbb{Z}_p$.
3. The last two digits of 7^{123} are 43.
4. For any sets A and B , if $X \subseteq A \times B$, then X can be written in the form $C \times D$ for some subsets $C \subseteq A$ and $D \subseteq B$.
5. For any map $f: X \rightarrow Y$ and any subsets $A, B \subseteq X$, we have $f(A \cap B) = f(A) \cap f(B)$. Here, $f(A) := \{f(a) : a \in A\}$.

Exercise 2.

- (a) Compute $5^{2025} \bmod 7$ (hint: write several first powers of 5 mod 7 and observe the pattern).
- (b) Compute $5^{2025} \bmod 11$.
- (c) Use the Chinese Remainder Theorem to find $5^{2025} \bmod 77$.

Exercise 3. A sequence (x_n) is defined by the rules

$$x_1 = 5, \quad x_{n+1} = 2x_n - 3.$$

Write down the first seven terms of the sequence. Try to guess the formula for x_n and prove it using induction in n .

Exercise 4. Show that if $n \geq 2$ and p is a prime that divides n but is such that p^2 is not a divisor of n , then

$$p^{\phi(n)+1} \equiv p \pmod{n}.$$

Here $\phi(n)$ is the Euler's phi-function.